



Applied Physics part-1

Applied Physics (Jawaharlal Nehru Technological University, Hyderabad)



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MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution – UGC, Govt.of India)

Recognizes under 2(f) and 12(B) of UGC ACT 1956

**(Affiliated to JNTUH, Hyderabad, Approved by AICTE –Accredited by NBA & NAAC-“A” Grade-ISO 9001:2015
Certified)**

APPLIED PHYSICS

B.Tech – I Year – I Semester

DEPARTMENT OF HUMANITIES AND SCIENCES



MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

B.TECH – I YEAR – I SEM – CSE/IT/ECE/EEE

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(R18A0011) APPLIED PHYSICS

COURSE OBJECTIVES:

1. To understand dual nature of the matter and behavior of a particle quantum mechanically.
2. To understand band structure of the solids and classification of materials.
3. To be able to distinguish pure, impure semiconductors and characteristics of PN junction diode.
4. To understand dielectric and magnetic properties of the materials and enable them to design and apply in different fields.
5. To be able to distinguish ordinary light with a laser light and realize the transfer of light through optical fibers.

UNIT – I

QUANTUM MECHANICS

(7 Hours)

Wave nature of particles, deBroglie's hypothesis, matter waves, Heisenberg's uncertainty principle, Davisson and Germer's experiment, G.P Thomson experiment, Schrodinger time-independent wave equation-significance of wave function, particle in one dimensional square well potential.

UNIT – II

ELECTRONIC MATERIALS

(7 Hours)

Free electron theory, Fermi level, Density of states, Periodic potential-Bloch's theorem, Kronig – Penny model, E – K diagram, Effective mass, Origin of energy bands in solids, Classification of materials on the basis of energy bands: Metals, semi conductors and insulators.

UNIT – III

SEMICONDUCTOR PHYSICS

(7 Hours)

Intrinsic and extrinsic semiconductors, Direct and indirect band gap semi conductors, Carrier concentration in intrinsic and extrinsic semi conductors. Dependence of Fermi level on carrier concentration and temperature, carrier transport: diffusion and drift, Formation of PN junction, V-I characteristics of PN diode, energy diagram of PN diode, Hall experiment, semiconductor materials for optoelectronic devices - LED, Solar cell.

UNIT-IV

DIELECTRICS AND MAGNETIC PROPERTIES OF MATERIALS

(10 Hours)

Dielectrics: Introduction, Types of polarizations (Electronic and Ionic) and calculation of their polarizabilities, internal fields in a solid, Clausius-Mossotti relation.

Magnetism: Introduction, Bohr magneton, classification of dia, para and ferro magnetic materials on the basis of magnetic moment, Properties of anti-ferro and ferri magnetic materials, Hysteresis curve based on domain theory, Soft and hard magnetic materials.

UNIT – V:

(9 Hours)

LASERS & FIBER OPTICS

Characteristics of lasers, Absorption, Spontaneous and Stimulated emissions, Einstein's Coefficients, Population inversion, Meta stable state, types of pumping, lasing action, Construction and working of Ruby Laser, Helium-Neon Laser, Applications of lasers.

Introduction to optical fiber, Construction and working principle of an Optical Fiber, Acceptance angle and Numerical aperture, Types of Optical fibers - Mode and Propagation through step and graded index fibers, Attenuation, Optical Fiber in Communication System, Applications of optical fibers.

COURSE OUTCOMES: After completion of studying Applied Physics the student is able to,

1. Know the basic principles of quantum mechanics and the importance of behavior of a particle.
2. Realize the importance of band structure of solids and their applications in various electronic devices.
3. Learn concentration estimation of charge carriers in semiconductors and working principles of PN diode.
4. Learn dielectric, magnetic properties of the materials and apply them in material technology.
5. Learn the principles and production of LASER beams and transfer of information by optical fiber communication systems.

TEXT BOOKS:

1. Engineering Physics by Arumugam, Anuradha publications.
2. Engineering Physics- B.K.Pandey, S.Chaturvedi, Cengage Learning.

REFERENCES:

1. Engineering Physics – R.K. Gaur and S.L.Gupta, Dhanpat Rai Publishers.
2. Engineering Physics, S Mani Naidu- Pearson Publishers.
3. Engineering physics 2nd edition –H.K.Malik and A.K. Singh.
4. Engineering Physics – P.K. Palaniswamy, Scitech publications.
5. Physics by Resnick and Haliday.

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IV	DIELECTRICS AND MAGNETIC PROPERTIES OF MATERIALS
V	LASERS & FIBER OPTICS

UNIT – I

QUANTUM MECHANICS

Classical Mechanics:-

It is the branch of physics, which explains the motion of macroscopic objects. It was introduced by sir Issac Newton.

Quantum Mechanics:-

It is the branch of physics which explains about the motion of microscopic particles like electrons, protons etc. It was introduced by Max Planck in 1990.

Waves & particles:

Wave:- A wave is nothing but spreading of disturbance in a medium. The characteristics/properties of waves are 1) Amplitude 2) Time period 3) Frequency 4) Wavelength 5) Phase 6) Intensity.

Particle:- A particle is a point in space which has mass & occupies space or region. The characteristics/properties of a particle are 1) Mass 2) velocity 3) Momentum 4) Energy etc.

Matter Waves or de-broglie-waves: The waves associated with a material particle are called as matter waves.

Difference between matter waves and electro-magnetic waves:-

Matter Wave	Electromagnetic waves(Light waves)
1. Matter wave is associated with moving particle or material particle.	1. Oscillating charged particle gives rise to EM wave.
2. Wavelength of matter wave is given as $\lambda = \frac{h}{mv}$	2. Wavelength of an electromagnetic wave is given by $\lambda = \frac{hc}{E}$
3. Wavelength of matter wave depends upon mass of the particle & velocity.	3. Wavelength of an electromagnetic wave depends upon the energy of the photon.
4. It can travel with a velocity greater than the velocity of light in vacuum.	4. It can travel with a velocity equal to the speed of light in vacuum i.e $c = 3 \times 10^8 \text{ m/s}$.
5. It is not a EM wave.	5. Electric field and Magnetic field oscillate perpendicular to each other & generate EM Waves.

de-broglie concept of dual nature of matter waves:-

In 1924, Louis de-broglie suggested that matter waves also exhibit dual nature like radiation(light). They are

- I. Wave nature
- II. Particle nature

Wave nature of matter waves is verified by Davisson & Germer experiment, G.P. Thomson experiment etc.

Particle nature of matter waves is verified by photo-electric effect, Compton effect etc.

de-broglie hypothesis:-

- 1) The universe consists of matter and radiation(light) only
- 2) Matter waves also exhibit dual nature like radiation.

- 3) The waves associated with the material particles are called as debroglie-waves or matter waves & the wave length associated with matter waves are called as de-broglie wave-length or matter wave-length (λ).
- 4) de-broglie wave-length is given by $\lambda = \frac{h}{p} = \frac{h}{mv}$

Expression for de-broglie wave-length(λ) in various form:-

According to the planck's theory of radiation, the energy of photon is given by

$$E = h\nu = \frac{hc}{\lambda} \text{----(1)}$$

h- planck's constant, ν -frequency of photon

According to Einstein mass energy relation

$$E = mc^2 \text{----(2)}$$

m-mass of a photon

c-velocity of light

From equation of (1) & (2)

$$\frac{hc}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mv} = \frac{h}{p} \text{--- (3)}$$

Where p-momentum of photon = mc

m-mass of photon, c-speed of light

But according to de-broglie theory

Momentum of electron particle(p) = mv

m-mass of e's, v-velocity of electron particle

$$\text{de-broglie wave-length}(\lambda) = \lambda = \frac{h}{mv} = \frac{h}{p} \text{----(4)}$$

Eq.(4) gives the expression for de-broglie wave-length.

Other forms of de-broglie wavelength(λ):-

(i) In terms of Energy(E):-

we know that the kinetic energy of particle i.e. $E = \frac{1}{2}mv^2$ --- (5)

Multiply Eq-(5) by 'm' on both sides, we get

$$mE = \frac{1}{2}mv^2$$

$$2Em = m^2v^2$$

$$mv = \sqrt{2mE}$$

$$\lambda = \frac{h}{\sqrt{2mE}} \text{----(6)}$$

(ii) de-broglie wavelength in terms of voltage(V):-

If a charged particle is accelerated through a potential difference(V), then the kinetic energy of the particle is given as

$$E = eV \text{---(7)}$$

But we have kinetic energy(E) of particle i.e. $E = \frac{1}{2}mv^2$

$$eV = \frac{1}{2}mv^2$$

$$2eV = mv^2$$

Multiply by 'm' on both sides we get

$$2meV = m^2v^2 \Rightarrow mv = \sqrt{2meV}$$

$$h = 6.6 \times 10^{-34} \text{ J-S}$$

$$\text{We have de-broglie wavelength } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2meV}} \text{ or } \lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$h = 6.625 \times 10^{-34} \text{ Js}$$

$$m_e = 9.1 \times 10^{-31} \text{ Kg}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

(iii) de-broglie wavelength in terms of Temperature(T):-

According to the kinetic theory of gases, the average kinetic energy of a particle at temperature 'T' is given by

$$E = \frac{3}{2} k_B T$$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m \times \frac{3}{2} k_B T}} = \frac{h}{\sqrt{3mk_B T}}$$

k_B is the Boltzmann constant.

Properties or characteristics of matter waves or de-broglie waves:-

- Lesser the mass of the particle, greater is the wavelength associated with it.
- Smaller the velocity of the particle, longer is the wavelength associated with the particle.
- When $V = 0$, $\lambda = \infty$ & $V = \infty$, $\lambda = 0$.
- Matter waves produced when the particles in motion are charged or uncharged.
- Matter waves are not electro-magnetic waves.
- Matter waves travel faster than the velocity of light.
- Wave nature of matter gives an uncertainty in the position of the particle.

Experimental Verification of matter waves:-

Here, we have two methods to verify the dual nature of matter waves. They are

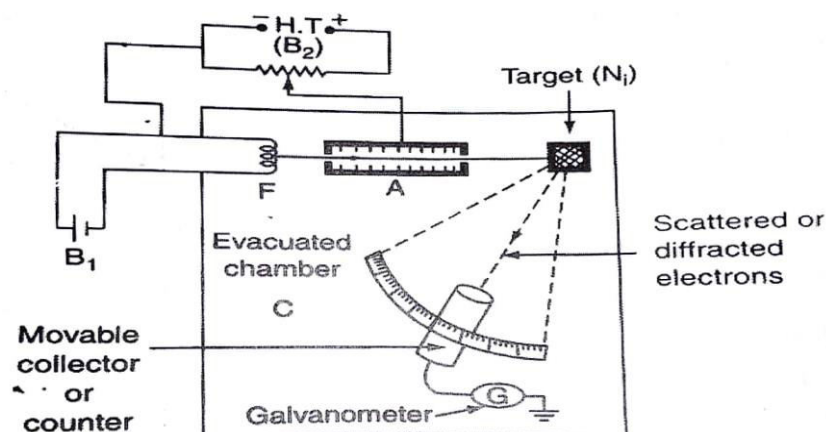
1. Davisson and Germer's experiment
2. G.P. Thomson's experiment

1. Davisson and Germer's experiment:-

First practical evidence for the wave nature of matter waves was given by C.J. Davisson and L.H. Germer in 1927. This was the first experimental support to de Broglie's hypothesis.

Principle: The e^- 's which are coming from the source are incident on the target and the e^- 's get diffracted. These diffracted e^- 's produce a diffraction pattern. It shows (explains) the wave nature of matter waves.

Experimental Arrangement:-



Construction:

The experimental setup is shown in above figure. It consists of mainly 3-parts

- (i) Filament
- (ii) Target
- (iii) Circular scale arrangement.

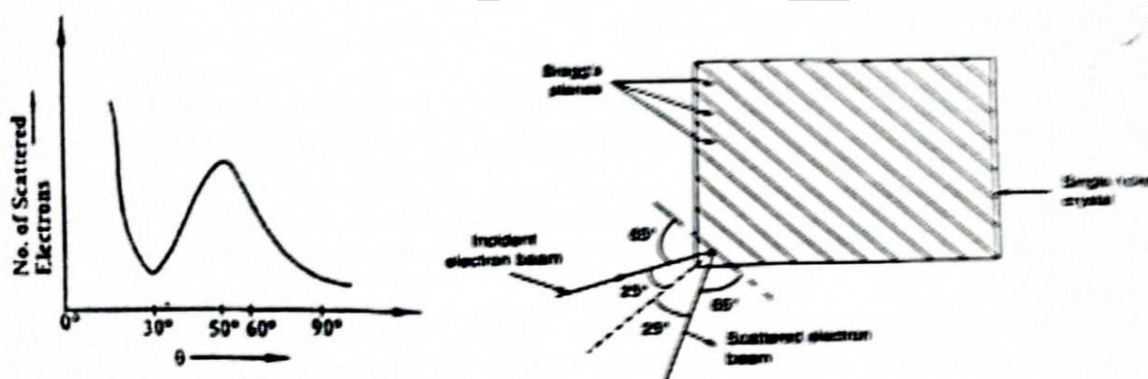
It also consists of a low tension battery(LTB), High tension battery(HTB) & a cylinder(A).

Working:

When tungsten filament 'F' is heated by a LTB then e's are produced. These e's are accelerated by High voltage(HTB). The accelerated e's are collimated into a fine beam of pencil by passing them through a system of pin-holes in the cylinder 'A'. This beam of electrons is allowed to incident on nickel crystal which acts as target. Then e's are scattered in all the directions. The intensity of scattered e's is measured by the circular scale arrangement. In this arrangement, an electron or movable collector(Double walled faraday cylinder) is fixed to circular scale which can collect the electrons and can move along the circular scale. The electron collector(Double walled faraday cylinder) is connected to a sensitive galvanometer to measure the intensity of electron beam entering the collector at different scattering angles(θ).

A graph is plotted between the scattering angle(θ) and the number of scattered electron's as shown in above figure.

The intensity of scattered e's is maximum at $\theta = 50^\circ$ & accelerating voltage =54V.

**Calculation of wave-length associated with e's:**

$$2d \sin \theta = n\lambda \text{---(1)}$$

For nickel crystal, $d = 0.909 \text{ \AA} = 0.909 \times 10^{-10} \text{ m}$

$n = 1$, First order

(i) Angle of diffraction(θ)(Practical value):

From the the figure,

$$180^\circ = \theta + \theta + 50^\circ$$

$$180^\circ = 2\theta + 50^\circ$$

$$\therefore \text{Diffraction angle}(\theta) = 65^\circ \text{----(2)}$$

Substituting the above values in eq – 1 we get

$$2 \times 0.909 \times \sin 65^\circ = 1 \times \lambda$$

$$\lambda = 1.65 \text{ \AA} \text{---(3)}$$

(ii) From de-broglie wave length(λ):

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$\text{But } V = 54\text{v}$$

$$\lambda = \frac{12.27}{\sqrt{54}} \text{ \AA} = 1.67 \text{ \AA}$$

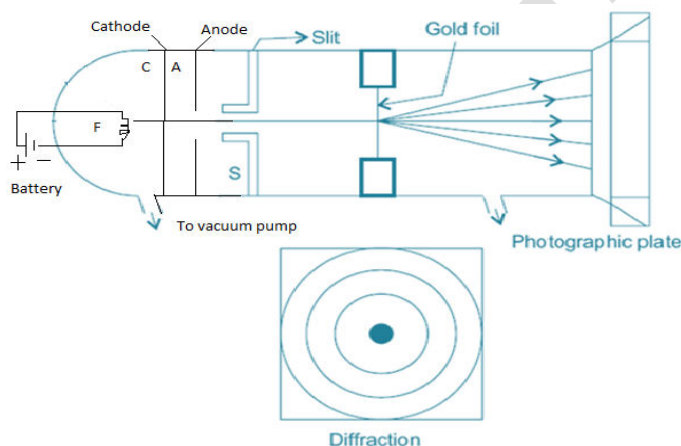
$$\lambda = 1.67 \text{ \AA} \text{ ----(4)}$$

From eq-(3) & eq-(4) it was been proved both the practical & theoretical wavelengths are almost equal. Hence the wave nature of particle is proved experimentally.

(2) G.P.Thomson experiment:

In 1928, G.P.Thomson, experimentally proved the dual nature of matter(particles).

Principle: The electrons which are coming from the source are incident on the thin metal foil e's are diffracted . These diffracted electrons produce a diffraction fringes on the photographic plate placed behind the foil. It explain the dual nature of matter.

Experimental Arrangement :

Construction: The experimental set up is shown in fig. It consists of

- (i) Filament(F)
- (ii) Anode(A)
- (iii) Cathode(C)
- (iv) Slit(S)
- (v) Gold foil(G)
- (vi) Photographic plate(P)
- (vii) Evacuated chamber.

Working:

Using a suitable battery the filament 'F' can be heated, so that electrons get emitted and pass through a high positive potential to the anode(A). Then 'e' beam passes through a slit incident on the gold foil(G) of thickness 10^{-8} cm.

The e's passing through the gold foil 'G' are recorded on a photographic plate(p). Since the gold foil consists of large no of micro crystals oriented at random ,the e's striking the gold foil diffracts according to bragg's law.

$$2d \sin \theta = n\lambda$$

After developing the plate, the diffraction pattern consists of concentric rings about a central spot is obtained. This pattern is similar as produced by x-rays.

To make sure this ,the pattern or fringes is deflected by application of magnetic field. Since the e's are deflected by magnetic field so that pattern is shifted. But x-rays are not deflected by electric & magnetic fields. So Thomson concluded that e's(matter) behave like waves.

Calculation's of wavelength of e's:

From fig, let us consider 'OA' is the radius(r) of the ring, 'O' is the center of the ring. Let 'L' be the distance between the gold foil(G) and the photographic plate(p).

Then from Δ^{le} QOA ,

$$\tan 2\theta = \frac{r}{L}$$

If 'θ' is small, then

$$\theta = \frac{r}{2L}$$

According to Bragg's law,

$$2d \sin \theta = n\lambda$$

If 'θ' is small, then $\sin \theta = \theta \Rightarrow 2d \cdot \theta = n\lambda$

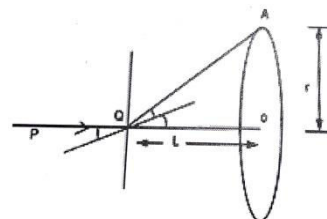
$$n\lambda = 2 \times d \times \frac{r}{2L}$$

$$n\lambda = \frac{dr}{L} \Rightarrow \lambda = \frac{dr}{nL} \text{---(1)}$$

de-Broglie wavelength of e's:

$$\therefore \lambda = \frac{12.27}{\sqrt{V}} \text{--- (2)}$$

eq-(1) & eq-(2) are almost equal. This confirms the existence of matter waves. The wave nature of matter(particles) is verified experimentally.

**Heisenberg uncertainty principle:-**

The discovery of dual nature of material particle imposes a serious hurdle in locating the exact position and momentum of a particle simultaneously.

This was removed by "Werner. Heisenberg" in 1927 by proposing a significant principle. Later it was called Heisenberg's uncertainty principle or Heisenberg's principle of indeterminacy.

"It is impossible to measure both the position and momentum of a particle simultaneously to any desired degree of accuracy." If 'Δx' & 'Δp' are uncertainties in the measurement of position & momentum of the particle then mathematically this uncertainty of these physical variables is written as

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \text{---(1)}$$

Similarly the uncertainties in measuring energy and time interval we can write

$$(i) \quad \Delta E \cdot \Delta t \geq \frac{h}{4\pi} \text{ and the uncertainties in measuring angular momentum \& angular displacement as}$$

$$(ii) \quad \Delta J \cdot \Delta \theta \geq \frac{h}{4\pi}$$

Explanation:-

(i) If $\Delta x = 0$ i.e., the position of a particle is measured accurately, then from eq-(1).

$$\Delta p = \frac{h}{\Delta x \cdot 4\pi}$$

$$\Delta p = \frac{h}{0} = \infty$$

It means that, the momentum of the particle can't be measured.

(ii) If $\Delta p = 0$ i.e., the momentum of a particle is measured accurately, from eq. (1).

$$\Delta x = \frac{h}{\Delta p \cdot 4\pi} = \frac{h}{0} = \infty$$

From the above said observations made by Heisenberg, he clearly states that it is impossible to design an experiment to prove the wave & particle nature of matter at any given instant of time.

If one measures position or momentum accurately, then there will be an uncertainty in the other. Thus, the Heisenberg's uncertainty principle gives the probability of determining the particle at any given instant of time in place of certainty.

Applications:-

- (i). It explains the non-existence of e's in the nucleus.
- (ii). It gives the binding energy of an e's in atom.
- (iii). It calculates the radius of Bohr's first orbit.

Schrodinger Wave Equation:-

Schrodinger describes the wave nature of a particle in mathematical form and is known as Schrodinger wave equation(SWE). There are two types of SWE

- (i). Schrodinger Time independent wave equation(STIWE)
- (ii). Schrodinger Time dependent wave equation(STDWE)

(i) Schrodinger Time independent wave equation(STIWE):-

To derive an expression for "(STIWE)" let us consider an electron (particle) moving along a positive direction along the axes. Let x, y, z be the coordinates of the particle & ' ψ ' is the wave-displacement or wave-function of the matter wave at any time 't'. It is assumed that ' ψ ' is finite, single-valued, continuous and periodic function.

We can express the classical differential wave equation of the material particle in three-dimension axes is given as

$$\frac{\partial^2 \psi}{\partial t^2} = v^2 \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right)$$

$$\text{In Three dimension we write } \frac{\partial^2 \psi}{\partial t^2} = V^2 \nabla^2 \psi \text{ --- (1)}$$

The solution of eq-(1) is given by

$$\psi = \psi_0 \sin \omega t \text{ --- (2)}$$

$$\psi = \psi_0 \sin 2\pi \nu t \text{ --- (3)}$$

Where $\omega = 2\pi \nu$ is the angular frequency of the particle

Differentiating eq-(3) w.r.t. 't' twice we get

$$\frac{\partial \psi}{\partial t} = \psi_0 \times \cos 2\pi \nu t \times 2\pi \nu$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\psi_0 \times \sin 2\pi \nu t \times 2\pi \nu \times 2\pi \nu$$

$$\frac{\partial^2 \psi}{\partial t^2} = -4\pi^2 \nu^2 \psi \text{ --- (3)}$$

$$\nu = \frac{c}{\lambda} = \frac{v}{\lambda}$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\frac{4\pi^2 v^2 \psi}{\lambda^2} \text{ --- (4)}$$

Substituting eq-(4) in eq-(1), we get,

$$V^2 \nabla^2 \psi = -\frac{4\pi^2 v^2 \psi}{\lambda^2}$$

$$\boxed{\nabla^2 \psi + \frac{4\pi^2}{\lambda^2} \psi = 0 \text{ --- (5)}}$$

According to de-broglie wave-length

$$\lambda = \frac{h}{p} = \frac{h}{mv} \text{ --- (6)}$$

Sub. eq(6) in eq(5), we get

$$\nabla^2 \psi + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \text{ --- (7)}$$

We have, the total energy (E) is given by

$$E = \text{P.E.} + \text{K.E.}$$

$$E = V + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = E - V$$

$$mv^2 = 2(E - V)$$

$$m^2v^2 = 2m(E - V) \text{ --- (8)}$$

Sub. eq-(8) in eq-(7) we get,

$$\nabla^2 \psi + \frac{4\pi^2}{h^2} 2m(E - V)\psi = 0$$

$$\nabla^2 \psi + \frac{8\pi^2 m (E - V)\psi}{h^2} = 0 \text{ --- (9)}$$

$$\nabla^2 \psi + \frac{2m(E-V)\psi}{\hbar^2} = 0 \text{ --- (10)}$$

$$\text{Where } \hbar = \frac{h}{2\pi}$$

Eq-(9) & Eq-(10) is called as Schrodinger time independent wave equation in three dimension

For a free particle, $V=0$,

$$\nabla^2 \psi + \frac{2mE\psi}{\hbar^2} = 0 \text{ --- (11)}$$

(ii) Schrödinger Time dependent wave equation:-

$$\hat{H}\psi = \hat{E}\psi \text{ where } \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V = \text{Hamiltonian operator}$$

$$\hat{E} = i\hbar \frac{\partial}{\partial t} = \text{Energy operator}$$

Physical significance of wave-function(ψ)(Eigen – function): –

Wave-function(ψ) or Eigen-function(ψ):-

It is a variable or complex quantity that is associated with a moving particle at any position (x,y,z) and at any time 't'.

(i) ' ψ ' of a particle is represented by $\psi = \psi_0 e^{-i\omega t}$

(ii) ' ψ ' explains the motion of microscopic particles.

(iii) ' ψ ' is a complex quantity & it does not have any meaning.

(iv) $|\psi|^2 = \psi\psi^*$ is real and positive, it has physical meaning.

(v) $|\psi|^2$ represents the probability of finding the particle per unit volume.

(vi) For a given volume $d\tau$, the probability of finding the particle is given by

$$\text{probability density}(p) = \iiint |\psi|^2 d\tau \text{ where } d\tau = dx dy dz$$

(vii) ' ψ ' gives the information about the particle behavior.

(viii) ' p ' values are between 0 to 1.

(xi) wave-function ' ψ ' is a single valued, finite and periodic function.

(x) If $p = \iiint |\psi|^2 d\tau = 1$, then ' ψ ' is called normalized wave function.

Application of Schrödinger Time independent wave equation(STIWE):-**(1) Particle or electron in a one dimensional box or particle in an infinite square well potential:-**

Consider a particle or electron of mass 'm' moving along x-axis enclosed in a one dimensional potential box as shown in figure. Since the walls are of infinite potential the particle does not penetrate out from the box. i.e. potential energy of the particle $V=\infty$ at the walls.

The particle is free to move between the walls A & B at $x=0$ and $x=L$.

The potential energy of the particle between the walls is constant because no force is acting on the particle.

∴ The particle energy is taken as zero for simplicity

i. e. $V = 0$ between $x=0$ & $x=L$.

Boundary Conditions:-

(i) The potential energy for particle is given as

$$V(x) = 0, \text{ for } 0 \leq x \leq L \quad \text{----- (1)}$$

$$V(x) = \infty, \text{ when } 0 \geq x \geq L$$

The Schrödinger time independent wave equation for the particle is given by,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V)\psi = 0$$

But $V=0$, between walls,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E)\psi = 0 \text{---- (2)}$$

$$\text{Let } \frac{8\pi^2m}{h^2} (E) = k^2$$

Then eq-(2), becomes

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \text{---- (3)}$$

The general solution of eq-(3) is given by

$$\psi(x) = A\sin kx + B\cos kx \text{----- (4)}$$

Where A, B are two constants, 'k' is the wave-vector

Applying the boundary conditions eq-(4), we get

$$(i) \quad \psi = 0 \text{ at } x = 0$$

$$0 = A\sin(0) + B\cos(0)$$

$$0 = 0 + B$$

$$B=0$$

Then eq-(4) is written as

$$\psi(x) = A\sin kx \text{ or } \psi_n(x) = A\sin(kx) \text{---- (6)}$$

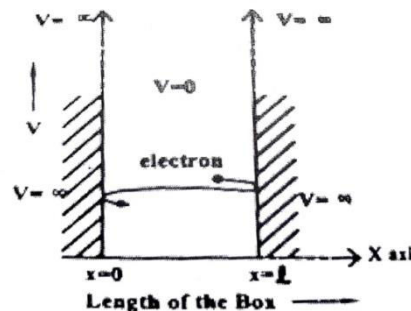
$$(ii) \quad \psi = 0 \text{ at } x = L$$

Eq-(6) can be written as

$$0 = A\sin(kL) \Rightarrow A \neq 0, \text{ and } \sin(kL) = 0$$

$$\sin(kL) = \sin(n\pi)$$

$$kL = n\pi \Rightarrow k = \frac{n\pi}{L} \text{----- (7)}$$



1) Energy of the particle(Electron):-

$$k = \frac{n\pi}{L} \Rightarrow k^2 = \frac{n^2 h^2}{L^2}$$

$$\text{But we have, } \frac{8\pi^2 m}{h^2} (E) = k^2$$

$$\frac{n^2 \pi^2}{h^2} = \frac{8\pi^2 m}{h^2} (E)$$

$$E = \frac{n^2 h^2}{8mL^2} \Rightarrow E_n = \frac{n^2 h^2}{8mL^2} \text{ --- (8)}$$

(ii) Energy levels of an particle(Electron):-

$$\text{when } n=1, \quad E_1 = \frac{h^2}{8mL^2}$$

$$\text{when } n=2, \quad E_2 = \frac{4h^2}{8mL^2} = 4E_1$$

$$\text{In general, } \boxed{E_n = n^2 E_1}$$

$\therefore$ Energy levels of an electron are discrete.

(iii) Normalization of the wave-function of the particle to find 'A':-

To find the 'A' value, by applying the normalization conditions.

$$\int_0^L |\psi(x)|^2 dx = 1$$

$$\int_0^L A^2 \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$$

$$A^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$$

By integrating and applying the appropriate limits, finally we obtain the value of 'A' as

$$A = \sqrt{\frac{2}{L}} \text{ --- (9)}$$

(iv) Electron wave functions ,probability density functions and energy functions:-

(i) Wave function of electron/particle

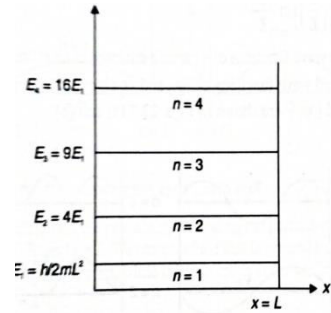
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \text{ --- (10)}$$

(ii) Probability density function(P)

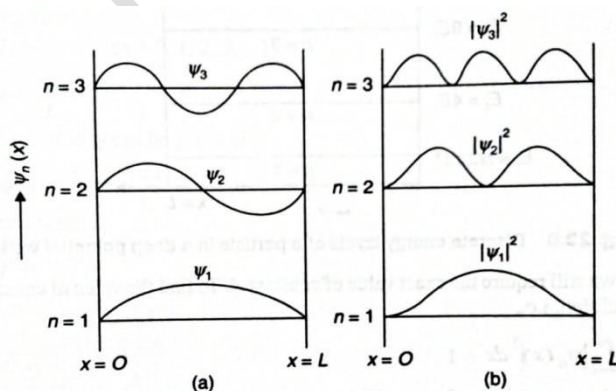
$$P = |\psi(x)|^2 = \frac{2}{L} \sin^2 \left(\frac{n\pi x}{L} \right) \text{ --- (11)}$$

(iii) Energy function of the particle

$$E_n = \frac{n^2 h^2}{8mL^2} \text{ --- (12)}$$



Discrete energy levels of a particle in a deep potential well



Wave function and probability density of a particle inside an infinite potential well

Unit II

ELECTRONIC MATERIALS

Free electron theory:

In solids, electrons in outer most orbits of atoms determine its electrical properties. Electron theory is applicable to all solids, both metals and non metals. In addition, it explains the electrical, thermal and magnetic properties of solids. The structure and properties of solids are explained employing their electronic structure by the electron theory of solids.

It has been developed in three main stages:

1. Classical free electron theory
 2. Quantum Free Electron Theory.
 3. Zone Theory.
- **Classical free electron theory:** The first theory was developed by Drude & Lorentz in 1900. According to this theory, metal contains free electrons which are responsible for the electrical conductivity and metals obey the laws of classical mechanics.
 - **Quantum Free Electron Theory:** In 1928 Sommerfield developed the quantum free electron theory. According to Sommerfield, the free electrons move with a constant potential. This theory obeys quantum laws.
 - **Zone Theory:** Bloch introduced the band theory in 1928. According to this theory, free electrons move in a periodic potential provided by the lattice. This theory is also called “Band Theory of Solids”. It gives complete informational study of electrons.

Classical free electron theory:

Even though the classical free electron theory is the first theory developed to explain the electrical conduction of metals, it has many practical applications. The advantages and disadvantages of the classical free electron theory are as follows:

Advantages:

1. It explains the electrical conductivity and thermal conductivity of metals.
2. It verifies ohm's law.
3. It is used to explain the optical properties of metals.
4. Metal composed of atoms in which electrons revolve around the nucleus are many states available for occupation. If the density of states is zero, no states can be occupied at that energy level.
5. The valence electrons are freely moving about the whole volume of the metals like the molecules of perfect gas in a container
6. The free electrons moves in random directions and collide with either positive ions or other free electrons. Collision is independent of charges and is elastic in nature
7. The movements of free electrons obey the laws of classical kinetic theory of gases
8. Potential field remains constant throughout the lattice.
9. In metals, there are large numbers of free electrons moving freely within the metal i.e. the free electrons or valence electrons are free to move in the metal like gaseous molecules, because nuclei occupy only 15% metal space and the remaining 85% space is available for the electrons to move.

Drawbacks:

1. It fails to explain the electric specific heat and the specific heat capacity of metals.
2. It fails to explain superconducting properties of metals.
3. It fails to explain new phenomena like photoelectric effect, Compton effect, black – body radiation, etc.
4. It fails to explain Electrical conductivity (perfectly) of semiconductors or insulators.
5. The classical free electron model predicts the incorrect temperature dependence of σ . According to the classical free electron theory, $\sigma \propto T^{-1}$.
6. It fails to give a correct mathematical expression for thermal conductivity.
7. Ferromagnetism couldn't be explained by this theory.
8. Susceptibility has greater theoretical value than the experimental value.

Quantum free electron theory of metals:**Advantages:**

1. All the electrons are not present in the ground state at 0 K, but the distribution obeys Pauli's exclusion principle. At 0 K, the highest energy level filled is called Fermi-level.
2. The potential remains constant throughout the lattice.
3. Collision of electrons with positive ion cores or other free electrons is independent of charges and is elastic in nature.
4. Energy levels are discrete.
5. It was successful to explain not only conductivity, but also thermionic emission paramagnetism, specific heat.

Drawbacks:

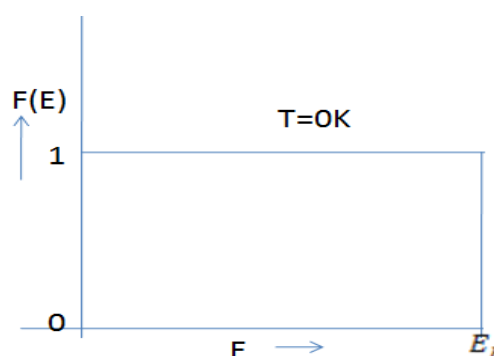
1. It fails to explain classification of solids as conductors, semiconductors and insulators.

Fermi level and Fermi energy:

The distribution of energy states in a semiconductor is explained by Fermi –Dirac statistics since it deals with the particles having half integral spin like electrons. Consider that the assembly of electrons as electron gas which behaves like a system of Fermi particles or fermions. The Fermions obeying Fermi –Dirac statistics i.e., Pauli's exclusion principle.

Fermi energy: It is the energy of state at which the probability of electron occupation is $\frac{1}{2}$ at any temperature above 0K. It separates filled energy states and unfilled energy states. The highest energy level that can be occupied by an electron at 0 K is called Fermi energy level

Fermi level: It is a level at which the electron probability is $\frac{1}{2}$ at any temp above 0K (or) always it is 1 or 0 at 0K.



Therefore, the probability function $F(E)$ of an electron occupying an energy level E is given by,

$$F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)} \dots \dots \dots (1)$$

Where E_F known as Fermi energy and it is constant for a system,

K is the Boltzmann constant and T is the absolute temperature.

Case I : Probability of occupation at $T=0K$, and $E < E_F$

Therefore $F(E) = 1$, as per above, clearly indicates that at $T=0K$, the energy level below the Fermi energy level E_F is fully occupied by electrons leaving the upper level vacant.

Therefore, there is 100% probability that the electrons to occupy energy level below Fermi level.

Case II: Probability of occupation at $T=0K$, and $E > E_F$

Then

i.e., all levels below E_F are completely filled and all levels above E_F are completely empty. As the temperature rises $F(E)$.

Case III: Probability of occupation at $T=0K$, and $E = E_F$

$$F(E) = \frac{1}{1 + e^0} = \frac{1}{1 + 1} = \frac{1}{2} = 0.5$$

The above condition states that, $T=0K$, there is a 50% probability for the electrons to occupy Fermi energy.

The probability function $F(E)$ lies between 0 and 1.

Hence there are three possible probabilities namely

$F(E) = 1$ 100% probability to occupy the energy level by electrons.

$F(E) = 0$ No probability to occupy the energy levels by electrons and hence, it is empty.

$F(E) = 0.5$ 50% probability of finding the electron in the energy level.

Density of States (DOS):

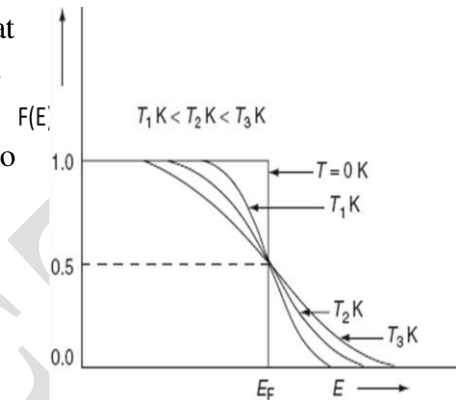
The number of electrons per unit volume in an energy level at a given temperature is equal to the product of density of states (number of energy levels per unit volume) and Fermi Dirac distribution function (the probability to find an electron).

$$n_c = \int g(E) \times f(E) dE \dots \dots \dots (1)$$

where n_c is the concentration of electrons, $g(E)$ is the density of states & $F(E)$ is the occupancy probability.

The number of energy states with a particular energy value E is depending on how many combinations of quantum numbers resulting in the same value n .

To calculate the number of energy states with all possible energies, we construct a sphere in 3D- space with 'n' as radius and every point (n_x , n_y and n_z) in the sphere represents an energy state.



As every integer represents one energy state, unit volume of this space contains exactly one state .Hence; the number of states in any volume is equal to the volume expressed in units of cubes of lattice parameters). Also $n^2 = n_x^2 + n_y^2 + n_z^2$

Consider a sphere of radius n and another sphere of radius $n+dn$ with the energy values are E and $(E+dE)$ respectively.

Therefore, the number of energy states available in the sphere of radius ' n ' is by considering one octant of the sphere (Here, the number of states in a shell of thickness dn at a distance ' n ' in coordinate system formed by n_x, n_y and n_z and will take only positive values ,in that sphere $\frac{1}{8}$ of the volume will satisfy this condition).

The number of energy states within a sphere of radius $(n+dn)$ is

$$\frac{1}{8} \left(\frac{4\pi}{3} \right) (n + dn)^3$$

Thus the number of energy states having energy values between E and $E+dE$ is given by

$$\begin{aligned} g(E)dE &= \frac{1}{8} \left(\frac{4\pi}{3} \right) (n + dn)^3 - \frac{1}{8} \left(\frac{4\pi}{3} \right) n^3 \\ &= \frac{1}{8} \left(\frac{4\pi}{3} \right) [(n + dn)^3 - n^3] \\ &= \frac{\pi}{6} (3n^2 dn) = \frac{\pi}{2} n^2 dn \end{aligned}$$

compared to ' dn ', dn^2 and dn^3 are very small .

Neglecting higher powers of dn

$$g(E)dE = \frac{\pi}{2} n^2 dn \dots \dots (2)$$

The expression for n^{th} energy level can be written as ,

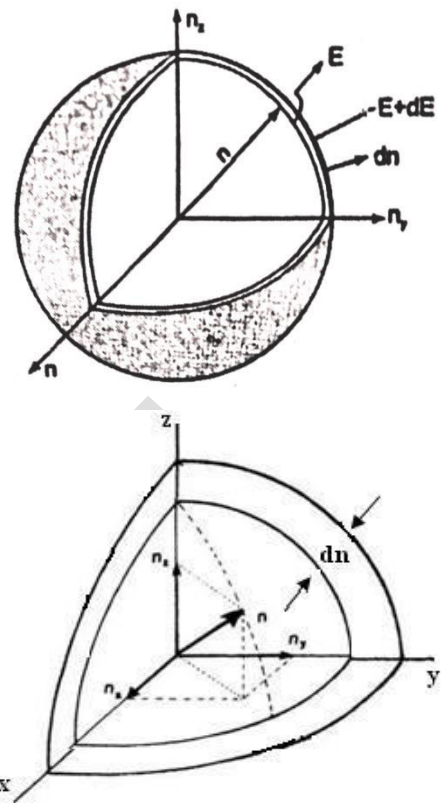
$$E = \frac{n^2 h^2}{8mL^2} \quad \text{or,} \quad n^2 = \frac{8mL^2 E}{h^2} \dots \dots (3)$$

$$\Rightarrow n = \left(\frac{8mL^2 E}{h^2} \right)^{\frac{1}{2}} \dots \dots (4)$$

Differentiating eq. (3):

$$2n dn = \frac{8mL^2}{h^2} dE \quad \Rightarrow \quad dn = \frac{1}{2n} \left(\frac{8mL^2}{h^2} \right) dE$$

$\therefore$ by substituting $1/n$ value in dn ,



$$n = \frac{1}{2} \left(\frac{8mL^2}{h^2} \right)^{\frac{1}{2}} \frac{dE}{E^{\frac{1}{2}}} \dots \dots \dots (5)$$

Substitute n^2 and dn from eq. (3) and (5), we get

$$g(E)dE = \frac{\pi}{2} \left(\frac{8mL^2}{h^2} \right) E dE \times \frac{1}{2} \left(\frac{8mL^2}{h^2} \right)^{\frac{3}{2}} \frac{dE}{E^{\frac{1}{2}}}$$

$$g(E)dE = \frac{\pi}{4} \left(\frac{8mL^2}{h^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}} dE \dots \dots \dots (6)$$

According to Pauli's Exclusion Principle, two electrons of opposite spin can occupy each energy state

Equation (6) should be multiplied by 2

$$g(E)dE = 2 \times \frac{\pi}{4} \left(\frac{8mL^2}{h^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}} dE$$

After mathematical simplification, we get g

$$g(E)dE = \frac{4\pi}{h^3} (2m)^{\frac{3}{2}} E^{\frac{1}{2}} L^3 dE$$

The density of energy states $g(E) dE$ per unit volume is given by,

$$g(E)dE = \frac{4\pi}{h^3} (2m)^{\frac{3}{2}} E^{\frac{1}{2}} dE \quad \because L^3 = 1$$

Bloch Theorem:

According to free electron model, a conduction electron in metal experiences constant potential. But in real crystal, there exists a periodic arrangement of positively charged ions through which the electrons move. As a consequence, the potential experienced by electrons is not constant but it varies with the periodicity of the lattice. In zone theory, as per Bloch, potential energy of electrons considered as varying potential with respect to lattice 'a'.

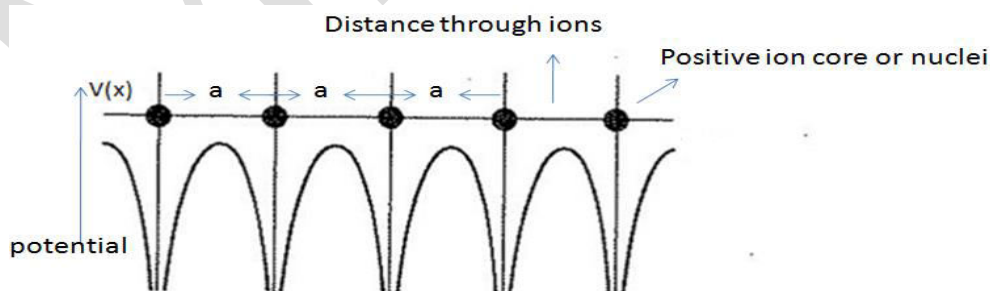


Fig: Variation of potential energy in a periodic lattice.

Let us examine one dimensional lattice as shown in figure. It consists of array of ionic cores along X-axis. A plot of potential V as a function of its position is shown in figure.

From graph:

At nuclei or positive ion cores, the potential energy of electron is minimum and in-between nuclei, the P.E. is considered as maximum w.r.to. Lattice constant 'a'.

This periodic potential $V(x)$ changes with the help of lattice constant a ,

$$V(x) = V(x + a) \quad ('a' \text{ is the periodicity of the lattice})$$

To solve, by considering Schrodinger's time independent wave equation in one dimension,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} [E - V(x)]\psi = 0 \dots \dots (1)$$

Bloch's 1D solution for Schrodinger wave equation (1) $\psi_k(x) = u_k(x)\exp(ikx) \dots \dots \dots (2)$

where $u_k(x) = u_k(x + a)$

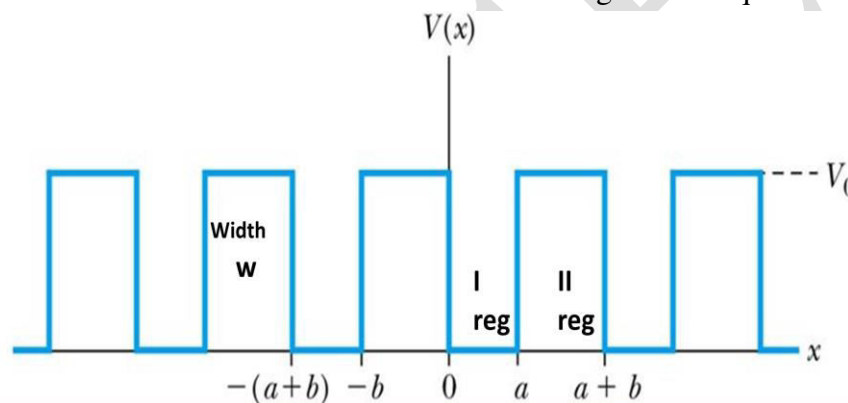
Here $u_k(x)$ -periodicity of crystal lattice, modulating function, k - propagation vector $= \frac{2\pi}{\lambda}$
 e^{ikx} is plane wave.

By applying eq.n (2) to eq.n (1), it is not easy to solve Schrodinger wave equation and Bloch cannot explain complete physical information about an e^- in periodic potential field. Then Kronig Penny model was adopted to explain the electrical properties of an e^- .

Kronig-Penney model:

Kronig-penny approximated the potentials of an e^- s inside the crystal in terms of the shapes of rectangular steps as shown, i.e. square wells is known as Kronig Penny model.

i.e. The periodic potential is taken in the form of rectangular one dimensional array of square well potentials and it is the best suited to solve Schrodinger wave equation.



It is assumed that the potential energy is zero when x lies between 0 and a , and is considered as I region. Potential energy is V_0 , when x lies between $-b$ and 0 . And considered as II region.

Boundary conditions:

$V(x) = 0$, where x lies between $0 < x < a$ - I region. $V(x) =$

V_0 , where x lies between $-b < x < 0$ - II region.

This model explains many of the characteristic features of the behavior of electrons in a periodic lattice.

The wave function related to this model may be obtained by solving Schrodinger equations for the two regions,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0, \text{ for } 0 < x < a \text{ with } V(x) = 0 \dots \dots \dots (1)$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi = 0, \text{ for } -b < x < 0 \text{ with } V(x) = V_0 \dots \dots \dots (2)$$

Again,

$$\frac{d^2\psi}{dx^2} + \alpha^2 \psi = 0 \dots\dots (3)$$

$$\text{where } \alpha^2 = \frac{2mE}{\hbar^2} \text{ and } \alpha = \frac{2\pi}{h} \sqrt{2mE}$$

$$\frac{d^2\psi}{dx^2} - \beta^2 \psi = 0 \dots\dots (4)$$

$$\text{where } \beta^2 = \frac{2m}{\hbar^2} (V_0 - E)$$

The solution of these equations from Bloch theorem, $\psi_k(x) = u_k(x)\exp(ikx)$. From figure, square well potentials, if V_0 increases, the width of barrier 'w' decreases, if V_0 decreases the width of barrier w increases. But the (product) barrier strength V_0w remains constant.

To get this, differentiating above Schrodinger wave equations 3 & 4 w.r.to x, and by applying boundary conditions of x (w.r.to their corresponding Ψ), to know the values of constants A, B of region -I, C,D-for reg-II, we get mathematical expression (by simplification)

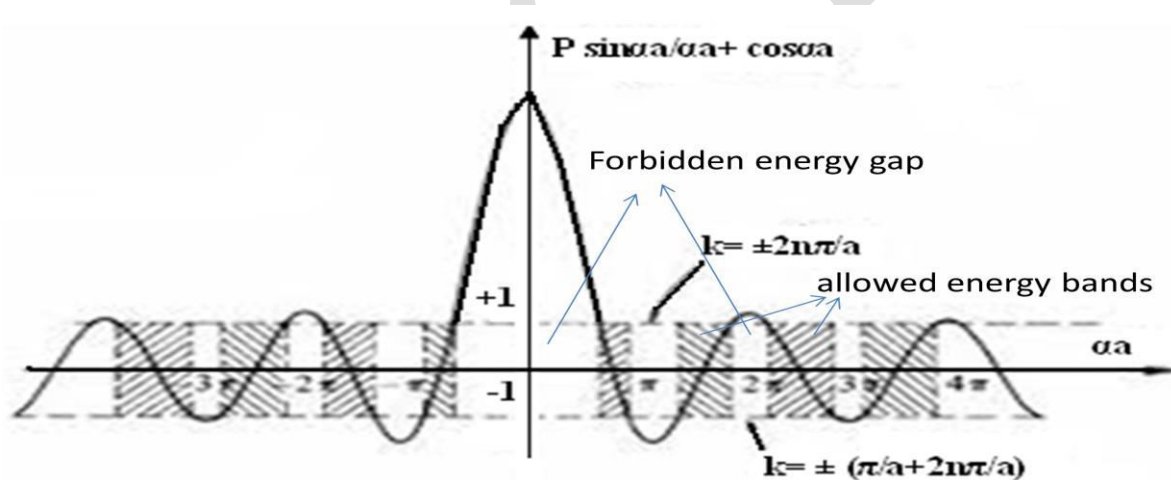
$$\cos ka = P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$$

where,

$$P = \frac{4\pi^2 ma}{h^2} V_0 w \quad \text{and} \quad \alpha = \frac{2\pi}{h} \sqrt{2mE}$$

P-varying term, known as scattering power.

And ' $V_0 w$ ' is known as barrier strength.



Conclusions:

1. The L.H.S is a cosine term which varies between the limits -1 and +1, and hence the R.H.S also varies between these limits. It means energy is restricted within -1 to +1 only.
2. If the energy of e^- lies between -1 to +1, are called **allowed energy bands** and it is shown by shaded portion in energy spectrum. This means that ' αa ' can take only certain range of values belonging to allowed energy band.
3. As the value of αa increases, the width of the allowed energy bands also increases.
4. If energy of e^- s not lies between -1 to +1 are known as **forbidden energy bands** and it decreases w.r.to increment of αa .
5. Thus, motion of e^- s. in a periodic lattice is characterized by the bands of allowed & forbidden energy levels.

Case 1:1. $P \rightarrow \infty$

If $P \rightarrow \infty$, the allowed band reduces to a single (line) energy level, gives us steeper lines.

We have

$$\cos ka \left(\frac{\alpha a}{p} \right) = \sin \alpha a + \cos \alpha a \left(\frac{\alpha a}{p} \right)$$

$$P \rightarrow \infty, \frac{1}{\infty} = 0 \quad \text{then} \quad \sin \alpha a = 0$$

$$\sin \alpha a = \sin n\pi$$

$$\alpha a = n\pi$$

$$\alpha^2 a^2 = n^2 \pi^2$$

$$\alpha^2 = \frac{n^2 \pi^2}{a^2}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{a^2}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{n^2 \pi^2 \hbar^2}{2ma^2 4\pi^2}$$

$$E = \frac{n^2 \hbar^2}{8ma^2} \quad , \text{ here } a \text{ is lattice constant}$$

It means, it (zone theory) supports quantum free electron theory.

Case 2:

$P \rightarrow 0$, We have

$$\cos ka = P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$$

$$\alpha a = ka$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{2mE}{\hbar^2} = \left(\frac{2\pi}{\lambda} \right)^2 = \frac{4\pi^2}{\lambda^2}$$

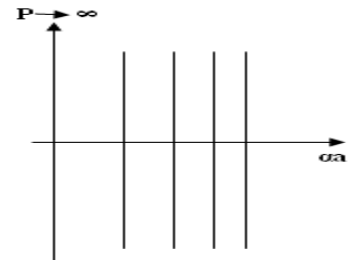
$$E = \frac{4\pi^2 \hbar^2}{2m\lambda^2}$$

$$E = \frac{4\pi^2 \hbar^2}{2m\lambda^2 4\pi^2}$$

$$E = \frac{1}{2} mv^2$$

It gives us kinetic energy of an electron. It means zone theory supports classical free electron theory at this situation electron completely free electron not bounded with allowed and forbidden gaps (and no energy level exists).

Thus by varying P from 0 to ∞ , we find that the completely free electron(s) becomes completely bound to Brillouin Zone.

Brillouin Zone OR E-K diagram:

The Brillouin zone are the boundaries that are marked by the values of wave vector k , in which electrons can have allowed energy values. These represent the allowed values of k of the electrons in 1D, 2D, & 3D.

We have, the energy of the electron in a constant potential box is,

$$E = \frac{n^2 h^2}{8ma^2} \dots (1) \text{ where } a = \text{length of the box.}$$

But,

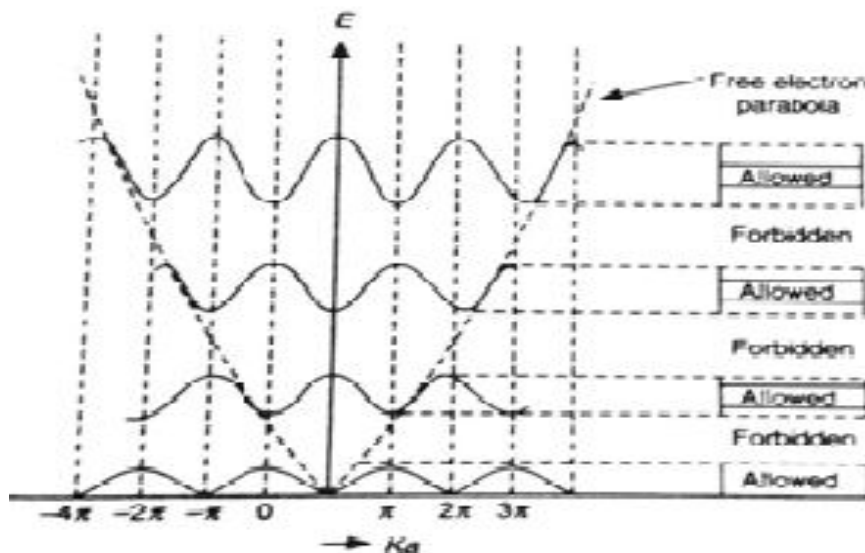
$$k = \frac{n\pi}{a} \Rightarrow k^2 = \frac{n^2 \pi^2}{a^2}$$

$$\frac{n^2}{a^2} = \frac{k^2}{\pi^2} \dots \dots \dots (2)$$

Substitute eqn (2) in (1), we get $E = \frac{k^2 h^2}{8m\pi^2}$; $E \propto k^2$. It represents parabolic equation.

A graph is drawn between the total energy (E) and the wave vector k , for various values of k .

i.e. $k = \frac{n\pi}{a}$; $n = \pm 1, \pm 2, \pm 3, \dots$



It is the energy spectrum of an electron moving in presence of a periodic potential field and is divided into allowed energy regions (allowed zones) or forbidden energy gaps (forbidden zones).

Allowed energy values lie in the region $k = -\pi/a$ to $+\pi/a$. This zone is called the first Brillouin zone. After a break in the energy values, called forbidden energy band, we have another allowed zone spread from $k = -\pi/a$ to $-2\pi/a$ and $+\pi/a$ to $+2\pi/a$. This zone is called the second Brillouin zone. Similarly, higher Brillouin zones are formed

Concept of effective mass of electron:

When an electron in a periodic potential of lattice is accelerated by an known electric field or magnetic field, then the mass of the electron is called effective mass and is represented by m^* .

To explain, let us consider an electron of charge 'e' and mass 'm' moving inside a crystal lattice of electric field E .

Then by taking known expression $F=ma$, can be considered here as $F = m * a \dots\dots(1)$

The acceleration $a = \frac{eE}{m}$ is not constant in the periodic lattice but varies due to the change in electronic mass.

If free electron under wave packet, the group velocity V_g corresponding to the particle's velocity can be written as

$$v_g = \frac{d\omega}{dk} = 2\pi \frac{dv}{dk} = \frac{2\pi}{h} \frac{dE}{dk} = \frac{1}{h} \frac{dE}{dk} \dots\dots(2) \text{ where } E = hv$$

The rate of change of velocity is known as

$$\text{Acceleration, } a = \frac{dv_g}{dt} = \frac{1}{h} \frac{d^2E}{dk dt}$$

$$a = \frac{1}{h} \frac{dE}{dt} \frac{dk}{dk} = \frac{1}{h} \frac{d^2E}{dk^2} \frac{dk}{dt} \dots\dots(3)$$

$$\text{From quantum mechanics relation, } p = \hbar k \dots\dots(4)$$

$$\text{and } F = \frac{dp}{dt} \dots\dots(5)$$

By differentiating eq (4) w.r.to t, and by substituting eq(5)

$$\hbar \frac{dk}{dt} = \frac{dp}{dt} = F \Rightarrow \frac{dk}{dt} = \frac{F}{\hbar} \dots\dots(6)$$

by substituting eq.(6) in eq (3),

$$\therefore a = \frac{1}{h} \frac{d^2E}{dk^2} \frac{F}{\hbar}$$

By rearranging the above term and by comparing with eq.(1)

$$F = \left[\frac{\hbar^2}{\left(\frac{d^2E}{dk^2} \right)} \right] a$$

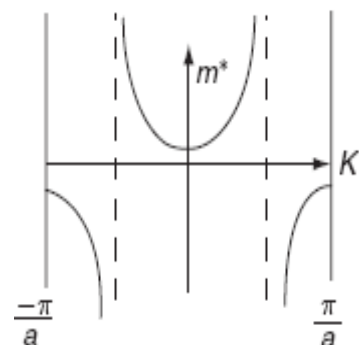
$$\therefore m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2} \right)}$$

Is known as expression for m^* and it is depends on E and K.

Variation of m^* with k:

The graph shows variation of m^* with k. Near $k=0$, effective mass approaches m . As the k value increases m^* increases, reaching its maximum value known as infinite effective mass.

Above the point of inflexion, m^* is negative and as k tends to $\frac{\pi}{a}$, it decreases to a small negative value called negative effective mass in the lower region. The positively charged particle which can be located in the lower region called negative effective mass



Origin of energy band formation in Solids:

The band theory of solids explains the formation of energy bands and determines whether a solid is a conductor, semiconductor or insulator.

The existence of continuous bands of allowed energies can be understood starting with the atomic scale. The electrons of a single isolated atom occupy atomic orbitals, which form a discrete set of energy levels.

When two identical atoms are brought closer, the outermost orbits of these atoms overlap and interact. When the wave functions of the electrons of different atoms begin to overlap considerably, the energy levels corresponding to those wave functions split.

If more atoms are brought together more levels are formed and for a solid of N atoms, each of the energy levels of an atom splits into N energy levels. These energy levels are so close that they form an almost continuous band. The width of the band depends upon the degree of overlap of electrons of adjacent atoms and is largest for the outermost atomic electrons.

As a result of the finite width of the energy bands, gaps are essentially leftover between the bands called forbidden energy gap.

The electrons first occupy the lower energy levels (and are of no importance) then the electrons in the higher energy levels are of important to explain electrical properties of solids and these are called valence band and conduction band.

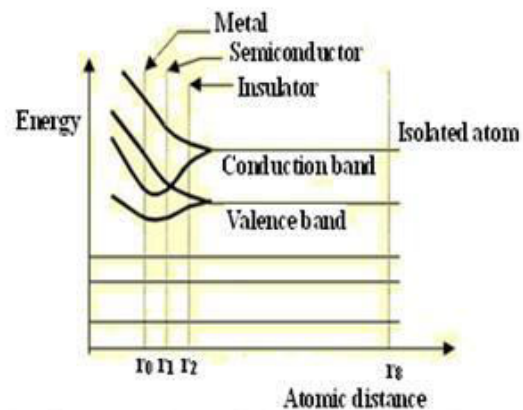
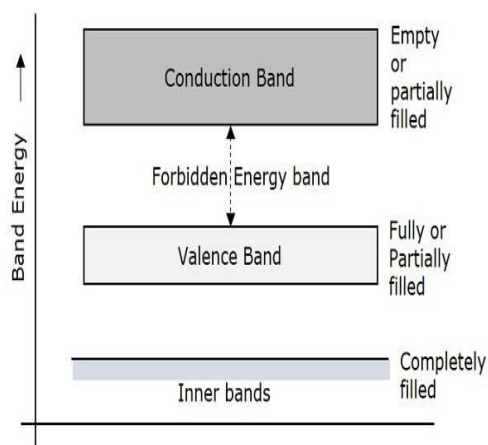


Fig: Formation of energy bands when atoms are closer

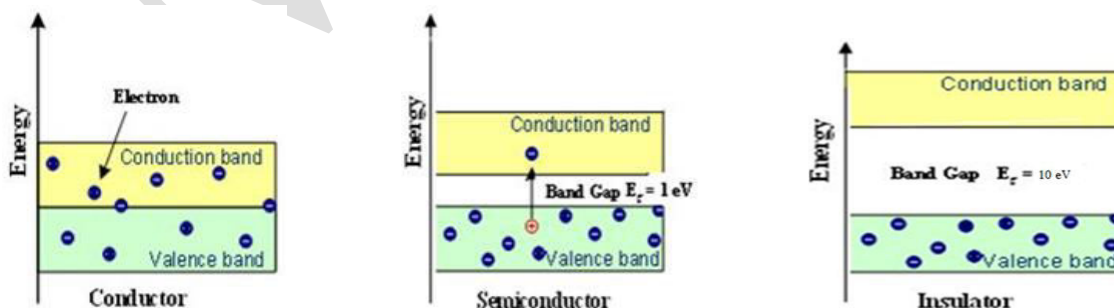
Valence band: A band occupied by valence electrons and is responsible for electrical, thermal and optical properties of solids and it is filled at 0K.

Conduction band: A band corresponding to outer most orbit is called conduction band and is the highest energy band and it is completely empty at 0K.

The **forbidden energy gap** between valence band conduction band is known as the energy band gap. By this solids are classified in to conductors, semiconductors and insulators.

Classification of solids into conductors , semiconductors & insulators:

Based on the energy band diagram materials or solids are classified as follows:



Conductors: In this kind of materials, there is no forbidden gap between the valence band and conduction band. It is observed that the valence band overlaps with the conduction band in metals as shown in figure. There are sufficient numbers of free electrons, available for electrical conduction and due to the overlapping of the two bands there is an easy transition

of electrons from one band to another band takes place, and there no chance for the presence of holes. Resistivity of conductors is very small and it is very few milli ohm meters. ($\Omega \text{ m}$).

Examples: All metals (Na, Mg, Al, Cu, Ni Cu, Ag, Li, Ar etc)

Semiconductors:

In semiconductors, there is a band gap exists between the valence band and conduction band and it is very less and it is the order of -1 to 2 eV are known as semiconductors. It will conduct electricity partially at normal conditions. The electrical resistivity values are 0.5 to 10^3 ohm meter. Due to thermal vibrations within the solid, some electrons gain enough energy to overcome the band gap (or barrier) and behave as conduction electrons. Conductivity exists here due to electronics and holes.

Examples: Silicon, Germanium, Ga As.

Insulators:

In insulators, the width of forbidden energy gap between the valence band and conduction band is very large. Due to large energy gap, electrons cannot jump from V.B to C.B. Energy gap is of the order of ~ 10 eV and higher than semiconductors. Resistivity values of insulators are 10^7 to 10^{12} ohm-m. Electrons are tightly bound to the nucleus, no valence electrons are available.

Examples: Wood, rubber, glass.

Unit III

SEMICONDUCTOR PHYSICS

SEMICONDUCTORS: The substances whose conductivity lies in between conductors and insulators are called as *semiconductors*.

Properties:

1. The conductivity of a semiconductors depends on it's temperature and it is increased with increase in temperature.
2. The energy gap of a semiconductor decreases with rise in temperature.
3. At 0 K a semiconductor becomes an insulator.
4. The absence of an electron in the VB of a semiconductor is called hole. Hole occur only in the VB. Or An empty electron state is called a hole.
5. Like electrons, the hole in the VB also conduct electricity in case of a semiconductor.
6. The electric current in a semiconductor is the sum of the currents due to electron and hole.
7. Hole is a positive charge carrier of current.

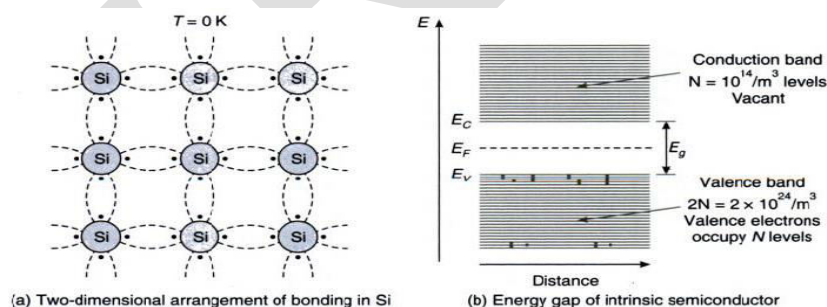
TYPES:

Depending on the relative concentration of holes and electrons the semiconductors are classified into two types.

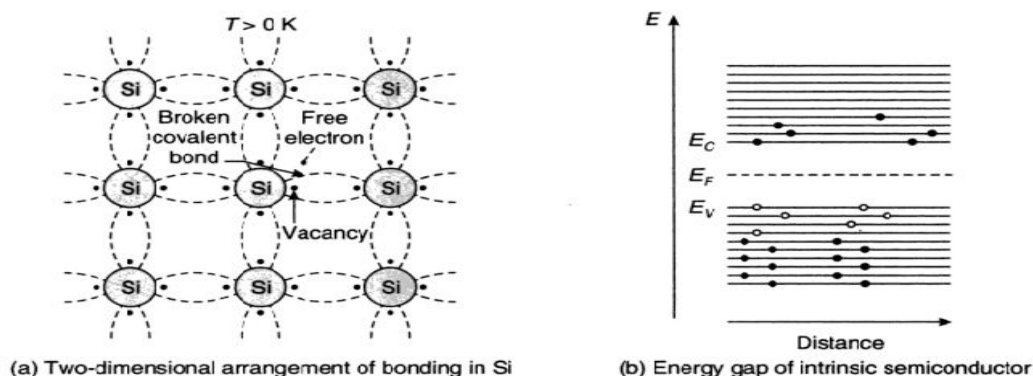
1. Intrinsic Semiconductor
2. Extrinsic Semiconductor.

Intrinsic Semiconductor: The semiconductor which is pure and having the number of electrons in CB equal to number of holes in VB is called as *intrinsic semiconductor*. The examples of intrinsic semiconductor are pure silicon and pure germanium crystals.

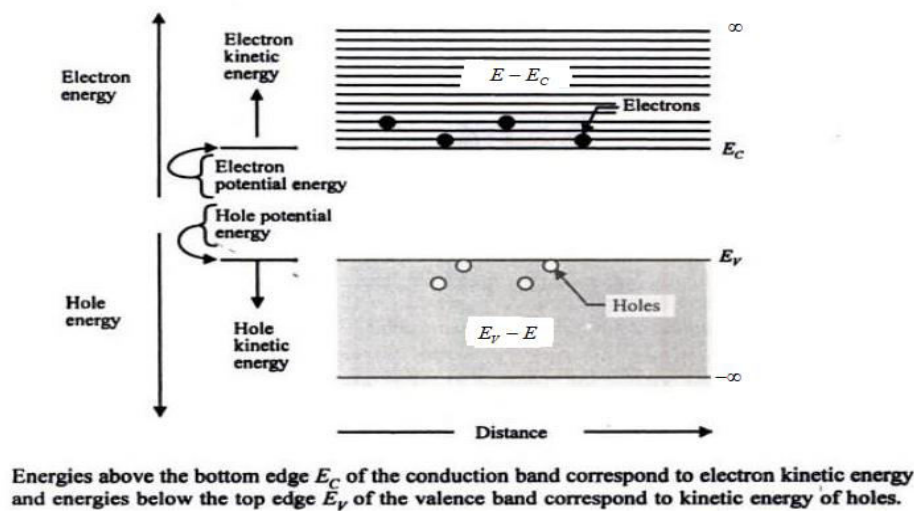
The intrinsic semiconductors at 0 K - The valence band is completely filled at 0 K and all the states in the conduction bands are vacant.



The intrinsic semiconductors at $T > 0\text{ K}$ - The valence band is not completely filled when $T > 0\text{ K}$ and some of the states in the conduction bands are occupied.



CARRIER CONCENTRATION IN AN INTRINSIC SEMICONDUCTOR:



The carrier concentration in an intrinsic semiconductor is determined by finding the concentration of electrons in the conduction band and the concentration of holes in the valence band and by applying the law of mass action.

At temperature above 0 K two types of free charge carriers exist in a semiconductor, electrons in the conduction band and holes in the valence band. In general, it is taken that the conduction band is extended from E_C to $+\infty$ and the valence band is extended from $-\infty$ to E_V .

Concentration of electrons in the conduction band:

In a semiconductor, the conduction band is completely empty and the valence band is completely filled at 0 K . If the temperature becomes greater than 0 K , some of the electrons in valence band get excited into the conduction band.

Let us calculate the electrons available in the conduction band, when $T > 0\text{ K}$. The electron concentration in the conduction band can be obtained using the equation

$$n = \int_{\text{Bottom of CB}}^{\text{Top of CB}} (\text{density of states})(\text{fermi function})$$

$$n = \int_{E_C}^{\infty} g(E)F(E) dE \quad (1)$$

Where $g(E) = \frac{4\pi}{h^3} (2m)^{3/2} E^{1/2}$ and $F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$

The electrons are moving in energy band but not in free space, so 'm' should be replaced by the effective mass (m_e^*). The term 'E' is replaced by $(E - E_C)$ because the bottom edge in the conduction band ' E_C ' is the potential energy of an electron at rest in the conduction band then $(E - E_C)$ represents the kinetic energy of the electron at high energy level.

$$n = \int_{E_C}^{\infty} \left[\frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_C)^{1/2} \right] \left[\frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)} \right] dE \quad (2)$$

Since, $E > E_F$, the term $\exp\left(\frac{E-E_F}{KT}\right) > 1$, Therefore, neglecting 1. Eq. (2) can be written as

$$n = \int_{E_C}^{\infty} \left[\frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_C)^{1/2} \right] \exp\left[-\left(\frac{E - E_F}{KT}\right)\right] dE \quad (3)$$

The above integral is solved by substituting $E - E_C = x$, then $dE = dx$, the lower limit of this integral becomes zero and the upper limit is ' ∞ '. Eq. (3) becomes

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_0^{\infty} [(x)^{1/2}] \exp\left[-\left(\frac{x + E_C - E_F}{KT}\right)\right] dx \quad (4)$$

Equation (4) can be written as

$$\begin{aligned} n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_0^{\infty} [(x)^{1/2}] \exp\left[-\left(\frac{E_C - E_F}{KT}\right)\right] \exp\left(\frac{-x}{KT}\right) dx \\ n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \exp\left(\frac{E_F - E_C}{KT}\right) \int_0^{\infty} (x)^{1/2} \exp\left(\frac{-x}{KT}\right) dx \end{aligned} \quad (5)$$

The integral $\int_0^{\infty} (x)^{1/2} \exp\left(\frac{-x}{KT}\right) dx$ can be evaluated by substituting $\frac{x}{KT} = y$
Then $KT dy = dx$. There is no change in the upper and lower limit values.

$$\begin{aligned} \int_0^{\infty} (x)^{1/2} \exp\left(\frac{-x}{KT}\right) dx &= \int_0^{\infty} (yKT)^{1/2} \exp(-y) (KT) dy \\ &= (KT)^{3/2} \int_0^{\infty} (y)^{1/2} \exp(-y) dy \end{aligned} \quad (6)$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \exp\left(\frac{E_F - E_C}{KT}\right) (KT)^{3/2} \int_0^{\infty} (y)^{1/2} \exp(-y) dy \quad (7)$$

The integral $\int_0^{\infty} (y)^{1/2} \exp(-y) dy$ can be solved using the Gamma function.

The Gamma function is

$$\begin{aligned} \int_0^{\infty} y^{(n-1)} \exp(-y) dy &= \Gamma(n) \\ \int_0^{\infty} y^{(3/2-1)} \exp(-y) dy &= \Gamma\left(\frac{3}{2}\right) \\ &= \Gamma\left(\frac{1}{2} + 1\right) \\ &= \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \end{aligned} \quad \boxed{\Gamma(n+1) = n \Gamma(n)} \quad \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$$

Substituting $\int_0^{\infty} y^{(3/2-1)} \exp(-y) dy$ in equation (7)

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \exp\left(\frac{E_F - E_C}{KT}\right) (KT)^{3/2} \frac{\sqrt{\pi}}{2}$$

$$n = 2 \left[\frac{2\pi m_e^* KT}{h^2} \right]^{3/2} \exp\left(\frac{E_F - E_C}{KT}\right)$$

$$\boxed{n = N_C \exp\left(\frac{E_F - E_C}{KT}\right)} \quad \text{Where } N_C = 2 \left[\frac{2\pi m_e^* KT}{h^2} \right]^{3/2}$$

The above eq. gives the electron concentration in the conduction band.

Concentration of holes in the valence band:

In a semiconductor the conduction band is completely empty and valence band is completely filled at **0 K**. If the temperature becomes greater than **0 K**, some of the electrons in valence band gets excited into the conduction band.

Let us calculate the number of holes available in the valence band, when **T > 0 K**. The concentration of holes in the valence band is calculated by

$$p = \int_{\text{Bottom of VB}}^{\text{Top of VB}} (\text{density of states})(\text{fermi function})$$

$$p = \int_{\text{Bottom of VB}}^{\text{Top of VB}} g(E) F_h(E) dE$$

We know that, the hole and electron occupation probability of the state of energy 'E' is always equal to 1.

It can be written as $F_e(E) + F_h(E) = 1$

The probability of occupancy of state by hole $F_h(E) = 1 - F_e(E)$

$$p = \int_{-\infty}^{E_V} g(E)[1 - F_e(E)] dE \quad (1)$$

Where $g(E) = \frac{4\pi}{h^3} (2m)^{3/2} E^{1/2}$ and $F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$

$$1 - F_e(E) = 1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$$

$$= \frac{\exp\left(\frac{E - E_F}{KT}\right)}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$$

Since, $E < E_F$, the term $\exp\left(\frac{E - E_F}{KT}\right)$ in the denominator is less than 1. So, it is negligible, Therefore

$$1 - F_e(E) = \exp\left(\frac{E - E_F}{KT}\right)$$

Substitute $g(E)$ and $1 - F_e(E)$ values in eq. (1)

$$p = \int_{-\infty}^{E_V} \left[\frac{4\pi}{h^3} (2m)^{3/2} E^{1/2} \right] \left[\exp\left(\frac{E - E_F}{KT}\right) \right] dE \quad (2)$$

The term E in eq. (2) is replaced by $(E_V - E)$ because E_V is the potential energy of a hole at rest. Then $(E_V - E)$ represents the kinetic energy of a hole at higher energy level. Note that the levels below the level E_F in the diagram are higher energy levels for holes. The hole is in energy band not in free space, so its mass is replaced by effective mass m_h^*

Then eq. (2) becomes

$$p = \int_{-\infty}^{E_V} \left[\frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_V - E)^{1/2} \right] \left[\exp\left(\frac{E - E_F}{KT}\right) \right] dE \quad (3)$$

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} \int_{-\infty}^{E_V} (E_V - E)^{1/2} \exp\left[-\left(\frac{E_F - E}{KT}\right)\right] dE \quad (4)$$

Let $E_V - E = x$, Then $-dE = dx$

The lower limit of this integral becomes ∞ and the upper limit is zero.

The integral (4) can be written as